

Chapter 3

Vectors, Norms, Dot Products, and Projections

Chapter overview

Chapter 2 introduced Euclidean space as the natural home for points such as (x, y) , (x, y, z) , and $(x_1, \dots, x_n)$. This chapter develops the algebra and geometry of **vectors** in that space.

A vector can represent a displacement, a velocity, a force, a row of data, a direction of change, a model parameter, or a feature embedding. In multivariable calculus, vectors are the language used to describe motion, tangent lines, tangent planes, gradients, flux, optimization, and linear approximation. In statistics and machine learning, vectors are also used to store observations, coefficients, residuals, gradients, and learned representations.

Main idea

A vector has both magnitude and direction. The norm measures magnitude, the dot product measures alignment, and projection separates a vector into a part along a chosen direction and a part perpendicular to that direction.

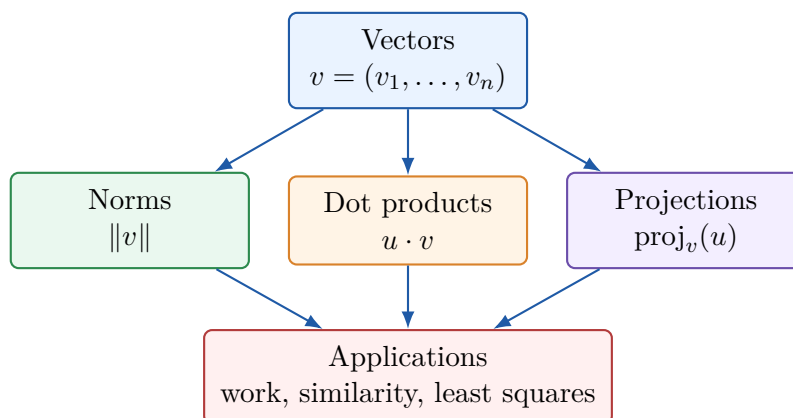


Figure 3.1: The main objects of the chapter and how they connect.

Learning goals

After studying this chapter, you should be able to:

1. perform vector addition, subtraction, and scalar multiplication in $\mathbb{R}^n$;
2. compute the Euclidean norm of a vector and construct unit vectors;
3. interpret distance as the norm of a displacement vector;
4. compute dot products and use them to find angles;
5. recognize orthogonal vectors and understand why orthogonality means zero dot product;
6. project one vector onto another and decompose a vector into parallel and perpendicular parts;
7. use the dot product to compute work and cosine similarity;
8. connect vector geometry to data science, statistics, optimization, and AI.

3.1 Vectors as displacement, direction, and data

Definition

A vector in $\mathbb{R}^n$ is an ordered list

$$v = (v_1, v_2, \dots, v_n).$$

Geometrically, we often draw a vector as an arrow. Algebraically, we compute with its components.

The same component list can represent different kinds of objects.

Context	Vector meaning
Geometry	displacement, direction, normal vector
Physics	force, velocity, acceleration
Statistics	data point, residual vector, coefficient vector
Machine learning	feature vector, weight vector, gradient vector
Optimization	search direction, update direction

If $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_n)$ are points, the displacement from A to B is

$$\overrightarrow{AB} = B - A = (b_1 - a_1, \dots, b_n - a_n).$$

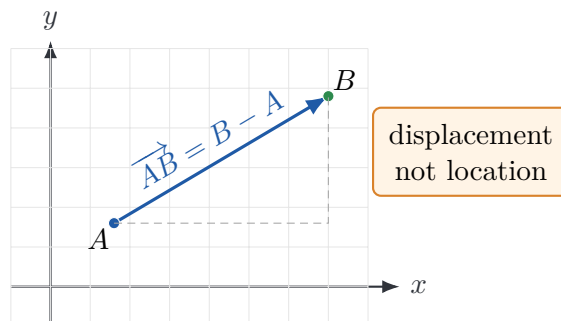


Figure 3.2: A displacement vector records how to move from one point to another.

Example 3.1: A displacement vector in space

Let

$$A = (1, -2, 3), \quad B = (2, 1, 5).$$

Then

$$\vec{AB} = (2 - 1, 1 - (-2), 5 - 3) = (1, 3, 2).$$

This vector tells us how to move from A to B : increase x by 1, increase y by 3, and increase z by 2.

Point versus vector

The ordered triple $(1, 3, 2)$ can be interpreted as a point or as a vector. As a point, it is a location. As a vector, it is a displacement. In calculus and applied mathematics, the interpretation matters as much as the arithmetic.

3.2 Vector addition and scalar multiplication

If

$$u = (u_1, \dots, u_n), \quad v = (v_1, \dots, v_n),$$

then vector addition is defined componentwise:

$$u + v = (u_1 + v_1, \dots, u_n + v_n).$$

If c is a real number, scalar multiplication is

$$cv = (cv_1, \dots, cv_n).$$

Geometrically:

- $u + v$ means follow displacement u and then displacement v ;
- $v - u$ means the displacement from the endpoint of u to the endpoint of v ;
- cv stretches v by the factor $|c|$;
- if $c < 0$, then cv also reverses direction.

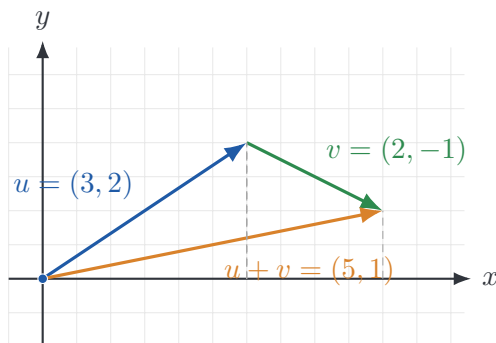


Figure 3.3: Vector addition by following one displacement after another.

Example 3.2: Net displacement

A particle is displaced first by $(3, 2)$ and then by $(2, -1)$. Its net displacement is

$$(3, 2) + (2, -1) = (5, 1).$$

The final location depends on the initial location, but the net displacement does not.

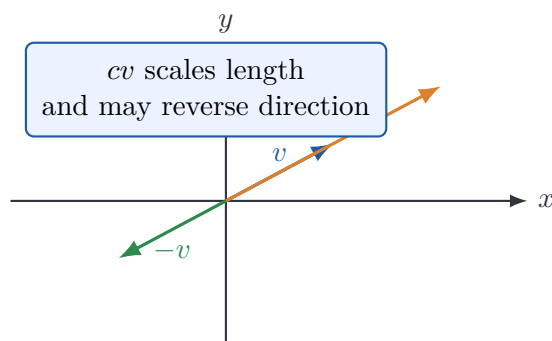


Figure 3.4: Scalar multiplication changes the magnitude of a vector, and negative scalars reverse its direction.

3.3 Norm, length, and unit vectors

Definition

The **Euclidean norm** of a vector $v = (v_1, \dots, v_n)$ is

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

A vector u is a **unit vector** if $\|u\| = 1$.

This generalizes the Pythagorean theorem. In $\mathbb{R}^2$, it gives the length of an arrow in the plane. In $\mathbb{R}^3$, it gives the length of an arrow in space. In $\mathbb{R}^n$, it gives the standard size of a data vector or parameter vector.

If $v \neq 0$, then the unit vector in the direction of v is

$$\hat{v} = \frac{v}{\|v\|}.$$

The vector v can be decomposed as

$$v = \|v\| \hat{v}.$$

This formula says that a nonzero vector is determined by two pieces of information: its magnitude and its direction.

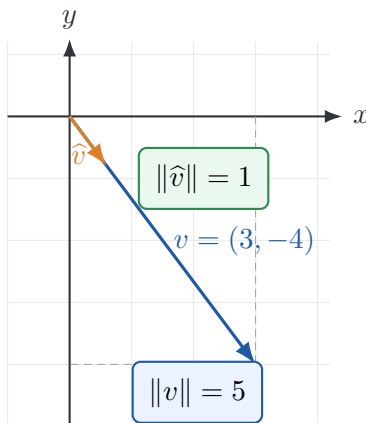


Figure 3.5: A vector and its unit direction have the same direction but different lengths.

Example 3.3: Unit vector parallel to $(3, -4)$

Let $v = (3, -4)$. Then

$$\|v\| = \sqrt{3^2 + (-4)^2} = 5.$$

Therefore a unit vector parallel to v and pointing in the same direction is

$$\hat{v} = \frac{1}{5}(3, -4) = \left(\frac{3}{5}, -\frac{4}{5}\right).$$

A unit vector parallel to v but pointing in the opposite direction is

$$-\hat{v} = \left(-\frac{3}{5}, \frac{4}{5}\right).$$

Example 3.4: Unit vectors perpendicular to $(3, -4)$

In the plane, if $v = (a, b)$, then $(-b, a)$ and $(b, -a)$ are perpendicular to v . For $v = (3, -4)$, two perpendicular vectors are

$$(4, 3), \quad (-4, -3).$$

Both have length 5, so the corresponding unit vectors are

$$\left(\frac{4}{5}, \frac{3}{5}\right), \quad \left(-\frac{4}{5}, -\frac{3}{5}\right).$$

Common mistake

Do not confuse a vector with its norm. The vector $(3, -4)$ has direction and magnitude. The norm $\|(3, -4)\| = 5$ is only a number.

3.4 Distance as norm of displacement

The distance between two points is the length of the displacement vector connecting them. If

$$A = (a_1, \dots, a_n), \quad B = (b_1, \dots, b_n),$$

then

$$\text{dist}(A, B) = \|B - A\| = \sqrt{(b_1 - a_1)^2 + \dots + (b_n - a_n)^2}.$$

This is one of the most important bridges between geometry and computation. In data science, distance is used in nearest-neighbor methods, clustering, anomaly detection, and similarity search.

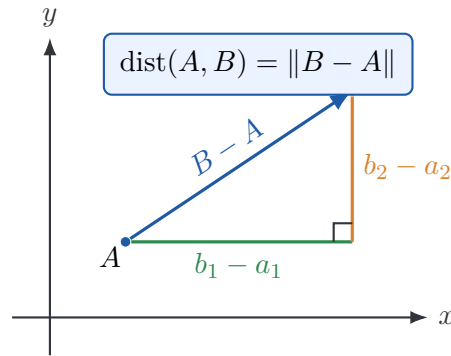


Figure 3.6: Distance is the norm of displacement. In two dimensions this is the Pythagorean theorem.

3.5 Dot product**Definition**

The **dot product** of two vectors

$$u = (u_1, \dots, u_n), \quad v = (v_1, \dots, v_n)$$

is

$$u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

The dot product takes two vectors and returns one number. It measures alignment.

The geometric identity is

$$u \cdot v = \|u\| \|v\| \cos \theta,$$

where θ is the angle between u and v . Therefore, for nonzero vectors,

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}.$$

This formula is central in geometry, statistics, and machine learning.

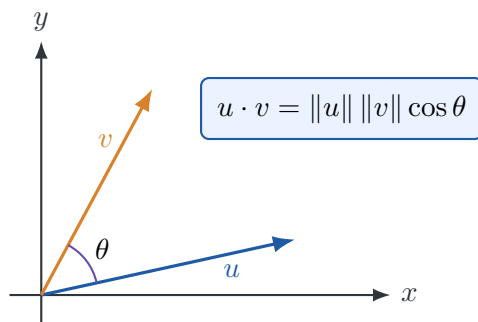


Figure 3.7: The dot product connects algebraic components with the angle between vectors.

Suppose u and v are nonzero.

- If $u \cdot v > 0$, the angle is acute and the vectors point generally in the same direction.
- If $u \cdot v = 0$, the angle is 90° and the vectors are orthogonal.
- If $u \cdot v < 0$, the angle is obtuse and the vectors point generally in opposite directions.

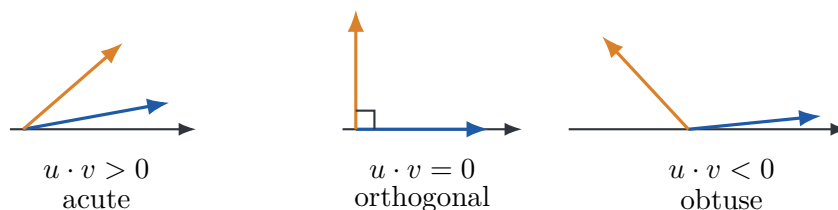


Figure 3.8: The sign of the dot product gives qualitative information about the angle.

Example 3.5: Angle between two vectors

Let

$$F = (1, 2), \quad v = (-1, 3).$$

Then

$$F \cdot v = (1)(-1) + (2)(3) = 5,$$

and

$$\|F\| = \sqrt{1^2 + 2^2} = \sqrt{5}, \quad \|v\| = \sqrt{(-1)^2 + 3^2} = \sqrt{10}.$$

Thus

$$\cos \theta = \frac{F \cdot v}{\|F\| \|v\|} = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{1}{\sqrt{2}}.$$

So

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}.$$

The angle is 45° , so the two vectors are strongly aligned.

3.6 Orthogonality

Definition

Two vectors u and v are **orthogonal** if

$$u \cdot v = 0.$$

Orthogonality generalizes perpendicularity. In $\mathbb{R}^2$ and $\mathbb{R}^3$, it means the vectors meet at a right angle. In $\mathbb{R}^n$, where we may not be able to draw the vectors, the equation $u \cdot v = 0$ is still meaningful.

Example 3.6: Checking orthogonality

Let

$$u = (2, -1, 3), \quad v = (1, 5, 1).$$

Then

$$u \cdot v = (2)(1) + (-1)(5) + (3)(1) = 2 - 5 + 3 = 0.$$

Therefore u and v are orthogonal.

Bridge to applications

Orthogonality is one of the main organizing ideas of applied mathematics. It appears in least squares, projections, Fourier series, PCA, QR factorization, residual analysis, and the geometry of independent directions in data.

3.7 Projection onto a direction

Projection answers a basic question:

How much of one vector points in the direction of another vector?

Let $v \neq 0$. The vector projection of u onto v is

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v.$$

The scalar component of u in the direction of v is

$$\text{comp}_v(u) = \frac{u \cdot v}{\|v\|}.$$

If $\hat{v} = v / \|v\|$ is the unit vector in the direction of v , then

$$\text{proj}_v(u) = (u \cdot \hat{v}) \hat{v}.$$

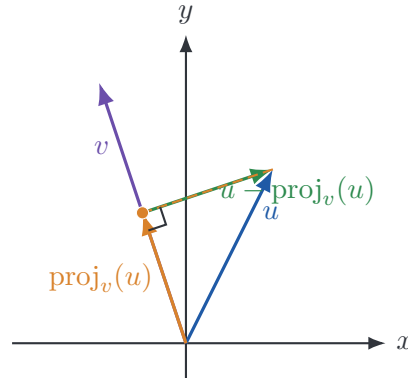


Figure 3.9: Projection decomposes a vector into a part parallel to v and a residual perpendicular to v .

Example 3.7: Projecting $(1, 2)$ onto $(-1, 3)$

Let

$$F = (1, 2), \quad v = (-1, 3).$$

Then

$$F \cdot v = 5, \quad v \cdot v = (-1)^2 + 3^2 = 10.$$

Therefore

$$\text{proj}_v(F) = \frac{F \cdot v}{v \cdot v} v = \frac{5}{10}(-1, 3) = \left(-\frac{1}{2}, \frac{3}{2}\right).$$

This is the part of F that lies in the direction of v .

Projection creates an orthogonal decomposition

Let

$$p = \text{proj}_v(u), \quad r = u - p.$$

Then

$$u = p + r,$$

where p is parallel to v and r is orthogonal to v .

To check this, compute

$$\begin{aligned} r \cdot v &= (u - p) \cdot v \\ &= u \cdot v - \left(\frac{u \cdot v}{v \cdot v} v \right) \cdot v \\ &= u \cdot v - \frac{u \cdot v}{v \cdot v} (v \cdot v) = 0. \end{aligned}$$

Bridge to least squares

In linear regression, the fitted values are projections of the response vector onto a column space. The residual vector is orthogonal to every column of the design matrix. This is why dot products and projections are essential for statistics.

3.8 Work as a dot product

If a constant force F moves an object through a displacement d , then the work done by the force is

$$W = F \cdot d.$$

Only the component of the force in the direction of motion contributes to work.

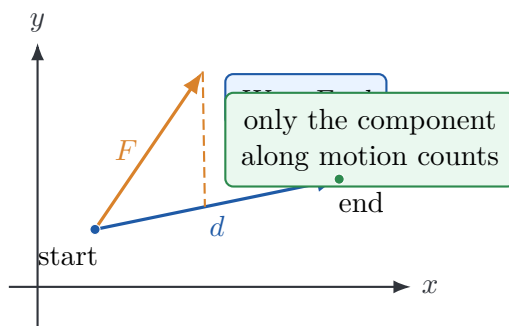


Figure 3.10: Work is the dot product of force and displacement.

Example 3.8: Work by a constant force

A constant force

$$F = (-1, -3, 4)$$

is applied while an object moves from

$$A = (1, 2, 0)$$

to

$$B = (2, 0, 1).$$

The displacement is

$$d = B - A = (1, -2, 1).$$

The work is

$$W = F \cdot d = (-1)(1) + (-3)(-2) + (4)(1) = -1 + 6 + 4 = 9.$$

The positive result means that the force has a net component in the direction of motion.

3.9 Cosine similarity and data vectors

For nonzero vectors u and v , the quantity

$$\frac{u \cdot v}{\|u\| \|v\|}$$

is the cosine of the angle between them. In data science and AI, this value is often called **cosine similarity**.

Cosine similarity focuses on direction rather than magnitude:

- similar direction: value near 1;
- orthogonal or unrelated direction: value near 0;
- opposite direction: value near -1 .

This is useful when the magnitude of a vector is less important than its pattern. For example, in text analysis, two documents may have different lengths but similar word-frequency directions.

Example 3.9: Cosine similarity of data vectors

Suppose two observations are

$$x = (2, 1, 0, 3), \quad y = (4, 2, 0, 6).$$

Then $y = 2x$, so the two observations point in exactly the same direction. Their cosine similarity is 1.

Now compare

$$x = (2, 1, 0, 3), \quad z = (0, 3, 4, 0).$$

The dot product is

$$x \cdot z = (2)(0) + (1)(3) + (0)(4) + (3)(0) = 3,$$

so the cosine similarity is positive but not close to 1.

	x	y	z	w
x	1.00	1.00	0.12	0.71
y	1.00	1.00	0.12	0.71
z	0.12	0.12	1.00	0.36
w	0.71	0.71	0.36	1.00

cosine similarity matrix

Figure 3.11: Cosine similarity measures direction rather than length. A value near 1 means similar direction.

3.10 High-dimensional geometry: most random directions are nearly orthogonal

High-dimensional geometry can be surprising. In low dimensions, random vectors often have visibly meaningful angles. In high dimensions, randomly chosen directions are usually close to orthogonal.

This fact helps explain why high-dimensional data can be difficult to visualize and why algorithms must be designed carefully.

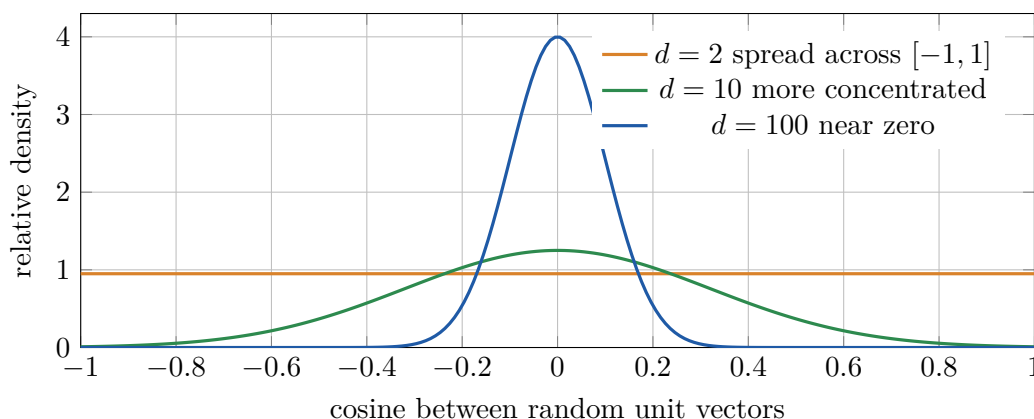


Figure 3.12: Cosines between random unit vectors concentrate near zero in high dimensions. The plotted curves are schematic density shapes emphasizing the concentration effect.

Modern interpretation

In high-dimensional data spaces, distance and angle can behave differently from our three-dimensional intuition. This is one reason why normalization, projection, regularization, and dimensionality reduction are central in statistics and machine learning.

3.11 Norms beyond the Euclidean norm

The Euclidean norm is the main norm used in this chapter, but other norms are important in applied mathematics and data science.

For $v = (v_1, \dots, v_n)$:

$$\|v\|_1 = |v_1| + \dots + |v_n|$$

is the ℓ^1 norm, and

$$\|v\|_\infty = \max\{|v_1|, \dots, |v_n|\}$$

is the ℓ^∞ norm. The Euclidean norm is

$$\|v\|_2 = \sqrt{v_1^2 + \dots + v_n^2}.$$

Different norms encode different ideas of size.

Norm	Formula	Common interpretation
$\ v\ _1$	sum of absolute values	sparsity, total variation, Manhattan distance
$\ v\ _2$	square root of sum of squares	Euclidean length, energy, least squares
$\ v\ _\infty$	maximum absolute component	worst-case component size

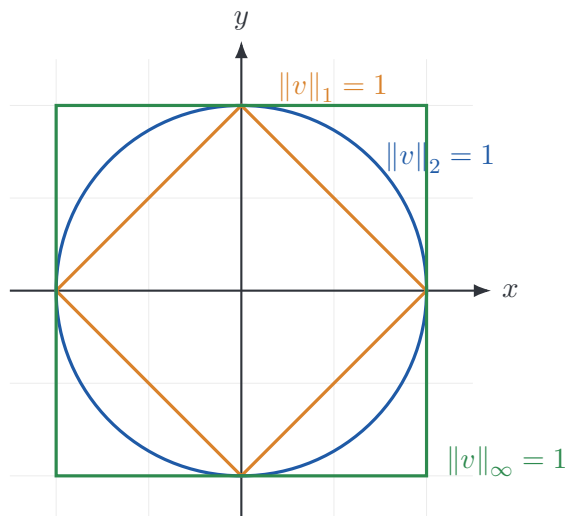


Figure 3.13: Different norms produce different unit balls in the plane.

Why this matters later

Least-squares regression is built from the ℓ^2 norm. Lasso regression is built from the ℓ^1 norm. Robust and worst-case methods often use ℓ^∞ -type constraints. Norms are not just geometric objects; they shape models.

3.12 Computational implementation of core vector operations

The formulas in this chapter are simple enough to compute by hand, but they also form the basic operations used by numerical linear algebra and machine learning libraries.

```

1 import numpy as np
2
3 u = np.array([1.0, 2.0])
4 v = np.array([-1.0, 3.0])
5
6 norm_u = np.linalg.norm(u)
7 norm_v = np.linalg.norm(v)
8 dot_uv = u @ v
9 cos_theta = dot_uv / (norm_u * norm_v)
10 projection = (dot_uv / (v @ v)) * v
11 residual = u - projection
12
13 print("||u|| =", norm_u)
14 print("||v|| =", norm_v)
15 print("u dot v =", dot_uv)
16 print("cos(theta) =", cos_theta)
17 print("projection of u onto v =", projection)
18 print("residual =", residual)
19 print("residual dot v =", residual @ v)

```

The last line should be approximately zero. This confirms that the residual $u - \text{proj}_v(u)$ is orthogonal to v .

3.13 Worked example summary

Example 3.10: Full vector analysis

Let

$$u = (2, -1, 2), \quad v = (1, 2, 2).$$

Step 1: Compute the norms.

$$\|u\| = \sqrt{2^2 + (-1)^2 + 2^2} = 3,$$

and

$$\|v\| = \sqrt{1^2 + 2^2 + 2^2} = 3.$$

Step 2: Compute the dot product.

$$u \cdot v = (2)(1) + (-1)(2) + (2)(2) = 2 - 2 + 4 = 4.$$

Step 3: Find the angle.

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|} = \frac{4}{9}, \quad \theta = \cos^{-1} \left(\frac{4}{9} \right).$$

Step 4: Project u onto v .

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v = \frac{4}{9} (1, 2, 2) = \left(\frac{4}{9}, \frac{8}{9}, \frac{8}{9} \right).$$

Step 5: Interpret. The vector u has a component in the direction of v and a residual component perpendicular to v . This is the same geometric idea that later appears in least-squares approximation.

3.14 Mini-project: cosine similarity for short documents

A simple way to represent a document is by a vector of word counts. Suppose our vocabulary is

(calculus, data, matrix, gradient).

Three short documents might have count vectors

$$d_1 = (3, 1, 0, 2), \quad d_2 = (2, 1, 0, 3), \quad d_3 = (0, 3, 4, 0).$$

Cosine similarity compares their directions.

```
1 import numpy as np
2
3 docs = np.array([
```

```

4     [3, 1, 0, 2],
5     [2, 1, 0, 3],
6     [0, 3, 4, 0]
7 ], dtype=float)
8
9 names = ["doc1", "doc2", "doc3"]
10 unit_docs = docs / np.linalg.norm(docs, axis=1, keepdims=True)
11 sim = unit_docs @ unit_docs.T
12
13 for i, name_i in enumerate(names):
14     for j, name_j in enumerate(names):
15         print(f"{name_i} vs {name_j}: {sim[i, j]:.3f}")

```

Documents d_1 and d_2 are more similar to each other than either is to d_3 . This is a small example of the geometry behind search, clustering, and embedding models.

3.15 Practice problems

Conceptual questions

1. Explain the difference between a point and a vector using an example in $\mathbb{R}^2$.
2. Why does multiplying a vector by a negative scalar reverse its direction?
3. What does it mean for two vectors in $\mathbb{R}^n$ to be orthogonal?
4. Why is the dot product a measure of alignment?
5. Explain why cosine similarity ignores magnitude.
6. Why is projection important for least-squares approximation?

Computational practice

1. Let $v = (6, 8)$. Find $\|v\|$ and a unit vector in the direction of v .
2. Let $u = (1, 2, 3)$ and $v = (4, 0, -1)$. Compute $u + v$, $u - v$, $3u$, and $-2v$.
3. Find the displacement vector from $A = (2, -1, 4)$ to $B = (5, 3, -2)$.
4. Compute the distance between $A = (2, -1, 4)$ and $B = (5, 3, -2)$.
5. Compute the dot product of $u = (2, -1, 3)$ and $v = (1, 5, 1)$. Are they orthogonal?
6. Find the angle between $u = (1, 1)$ and $v = (1, -1)$.
7. Find the projection of $u = (3, 4)$ onto $v = (1, 0)$.
8. Find the projection of $u = (1, 2)$ onto $v = (-1, 3)$.
9. A force $F = (2, -1, 4)$ moves an object from $A = (0, 1, 2)$ to $B = (3, 0, 5)$. Compute the work done.
10. Let $x = (1, 0, 2, 2)$ and $y = (2, 0, 4, 4)$. Compute their cosine similarity.

Applied and data-oriented problems

1. Suppose two students have feature vectors $x = (80, 90, 70)$ and $y = (40, 45, 35)$ representing scores on three assessments. Compute their cosine similarity. Interpret why this similarity may be high even though the magnitudes differ.
2. Suppose $r = y - \hat{y}$ is a residual vector in a regression problem. What does it mean geometrically if r is orthogonal to the fitted-value vector $\hat{y}$?
3. In a nearest-neighbor method, why might scaling the features change the computed distances?
4. Find a vector perpendicular to $(5, -2)$. Then normalize it.
5. Let $w = (w_1, w_2, \dots, w_n)$ be a model parameter vector. Explain why $\|w\|_2$ can be interpreted as a measure of model size.

Challenge problems

1. Prove that if $v \neq 0$, then $u - \text{proj}_v(u)$ is orthogonal to v .
2. Prove the Cauchy-Schwarz inequality in the special case of $\mathbb{R}^2$:

$$|u \cdot v| \leq \|u\| \|v\|.$$

3. Show that the projection formula minimizes the distance from u to the line spanned by v :

$$\min_{c \in \mathbb{R}} \|u - cv\|.$$

4. Let u and v be nonzero vectors. Show that their cosine similarity is unchanged if u and v are multiplied by positive constants.
5. In $\mathbb{R}^n$, show that if u is orthogonal to $v_1, \dots, v_k$, then u is orthogonal to every linear combination $c_1 v_1 + \dots + c_k v_k$.